Agenda

Today

- Short review of Lab 2
- Short review of point estimation; Class exercise
- Maximum likelihood estimators
- Probability plots
- Central Limit Theorem (CLT)

LearnHub:

- Lecture notes
- Class exercises and their solutions
- Reading assignment 1: Chapter 1 (Assessments) due next Monday.
- Lab 3: today

Point estimators

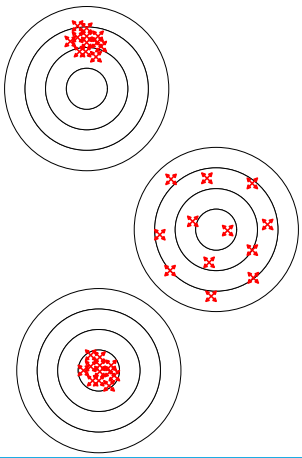
Let $X_1,\ldots,X_n$ be a (simple) random sample with a common probability distribution $f_X(x;\theta)$, where θ is the parameter of the distribution.

Let $\widehat{\Theta} = h(X_1,\ldots,Xn)$, which is a statistic. $\widehat{\Theta}$ is a random variable with a distribution that is a function of θ.

A statistic $\widehat{\Theta}$ that is used to estimate the value of a parameter θ is called point estimator.

The distribution of $\widehat{\Theta}$ depends on the distribution $f_X(x;\theta)$, so it depends on the parameter θ. Although θ is unknown, we can determine the distribution of $\widehat{\Theta}$ or the expected value of $\widehat{\Theta}$ or the variance of $\widehat{\Theta}$ in terms of θ.

Point Estimation



Target → Population parameter θ
(such as mean, variance, median etc.)

Shot → Point estimate of a parameter θ
(a sensible value for θ, a single number)

Shooter → Point estimator of a parameter θ
(a rule, mostly a formula, to calculate an estimate based on the sample)

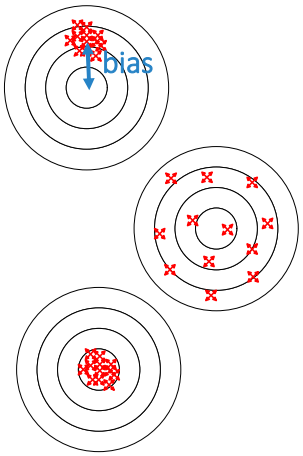
Mean Square Error of an Estimator

Definition

The **mean square error** of an estimator $\hat{\Theta}$ of the parameter θ is defined as

$$\mathrm{MSE}(\hat{\Theta}) = E(\hat{\Theta} - \theta)^2 \qquad (7\text{-}3)$$

Mean Square Error of an Estimator



$$\mathrm{MSE}_{\hat{\theta}} = \underbrace{\left(\mu_{\hat{\theta}} - \theta\right)^2}_{\text{Bias of } \hat{\theta}} + \underbrace{\left(\sigma_{\hat{\theta}}\right)^2}_{\text{standard error of } \hat{\theta}}$$

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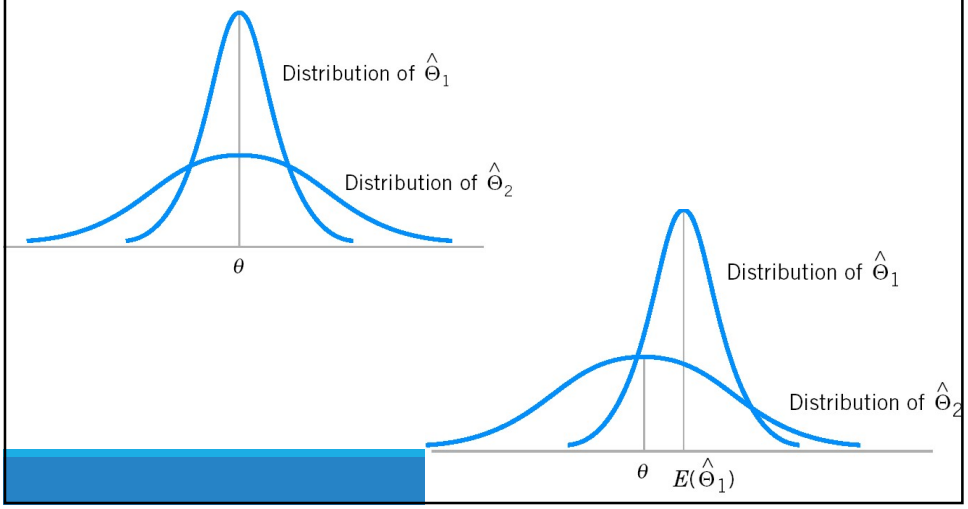
Unbiased estimators

Definition: A point estimator $\hat{\theta}$ is said to be an unbiased estimator of θ if $\mathrm{E}[\hat{\theta}] = \theta$ for all possible values of θ.

If $\hat{\theta}$ is not unbiased, $\mathrm{E}[\hat{\theta}] - \theta$ is called the bias of $\hat{\theta}$.

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General Concepts of Point Estimation



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Class exercise

Let X_1 and X_2 be two independent and identical random variables with mean μ and standard deviation σ. We define the following random variables:

$$\hat{\Theta}_1 = \frac{2X_1+X_2}{4} \quad \text{and} \quad \hat{\Theta}_2 = \frac{X_1+3X_2}{4}$$

a) Find the expected value of $\hat{\Theta}_1$ and $\hat{\Theta}_2$ in terms of mean μ and standard deviation σ.
b) Find the variance of $\hat{\Theta}_1$ and $\hat{\Theta}_2$ in terms of mean μ and standard deviation σ.
c) If $\hat{\Theta}_1$ and $\hat{\Theta}_2$ are used as estimators of μ, are they biased or unbiased estimators? Why or why not? Explain.
d) If you are supposed to choose one of them as a better estimator of μ, which one will you choose? Why or why not? Explain.

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What else is unbiased?

Proposition: Let X be a random variable with mean μ & variance σ^2. Let $X_1, \ldots, X_n$ be a random sample of size *n* from X.

b. S^2 is an unbiased estimator of σ^2.

$$S^2 = \frac{\sum_{i=1}^{n}(X_i - \bar{X})^2}{n-1} = \frac{1}{n-1}\left(\sum_{i=1}^{n} X_i^2 - \frac{(\sum_{i=1}^{n} X_i)^2}{n}\right)$$

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$$S^2 = \frac{\sum_{i=1}^{n}(X_i - \bar{X})^2}{n-1}$$

Example

We can show that $E[S^2] = \sigma^2$ and $\text{Var}(S^2) = \frac{2\sigma^4}{n-1}$.

Let $\hat{\sigma}^2 = \frac{\sum_{i=1}^{n}(X_i - \bar{X})^2}{n}$. Find MSE of $\hat{\sigma}^2$.

$$\text{MSE}_{\hat{\theta}} = \left(\mu_{\hat{\theta}} - \theta\right)^2 + \left(\sigma_{\hat{\theta}}\right)^2$$

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How to find good estimators?

Maximum likelihood estimators

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Example:

A random sample of 10 new bike helmets manufactured by a certain company is obtained. Assume that we would like to estimate the probability p = P{flawed helmet}.

$$X_i = \begin{cases} 1 & \text{if helmet is flawed} \\ 0 & \text{otherwise} \end{cases}$$

Observed X_i's = (1, 0, 1, 0, 0, 0, 0, 0, 0, 1)

What is the probability of observing this sequence in terms of p,

i.e., $L(X_1, \ldots, X_m; p)$=?

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Methods of Maximum Likelihood

Suppose that X is a random variable with probability distribution $f(x; \theta)$, where θ is a single unknown parameter. Let $x_1, x_2, \ldots, x_n$ be the observed values in a random sample of size n. Then the **likelihood function** of the sample is

$$L(\theta) = f(x_1; \theta) \cdot f(x_2; \theta) \cdot \cdots \cdot f(x_n; \theta) \qquad (7\text{-}5)$$

Note that the likelihood function is now a function of only the unknown parameter θ. The **maximum likelihood estimator** of θ is the value of θ that maximizes the likelihood function $L(\theta)$.

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Example:

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$$X_i = \begin{cases} 1 & \text{if helmet is flawed} \\ 0 & \text{otherwise} \end{cases}$$

Observed X_i's = (1, 0, 1, 0, 0, 0, 0, 0, 0, 1)

What is the probability of observing this sequence in terms of p, or the likelihood function $L(X_1, \ldots, X_m; p)$=?

Find $\hat{p}$ for which $L(X_1, \ldots, X_m; p)$ is maximized.

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Example

Sample: $X_1, \ldots, X_n$, where X_i's are normally distributed. Assume that σ^2 is known.

Find the maximum likelihood estimator of μ.

$$f(x;\mu,\sigma^2) = \frac{1}{\sqrt{2\pi}\sigma} e^{-(x-\mu)^2/2\sigma^2}$$

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A Desirable Property of the MLE

Proposition: Under very general conditions on the joint distribution of the sample, when the sample size n is large, the mle of any parameter θ:

- is approximately unbiased, i.e., $E(\hat{\theta}) \approx \theta$
- has a variance that is nearly as small as can be achieved by any estimator, i.e, the MLE is approximately the Minimum Variance Unbiased Estimator (MVUE) of θ.

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